MASTER'S THESIS · MSC MATHEMATICS

VU VRIJE UNIVERSITEIT AMSTERDAM

Exact generating-function analysis of a two-class priority M/M/1 queue

with jockeying and abandonment

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Waiting lists that change while you wait

In elderly-care admission, medical severity sets a customer's **priority class**. But waiting is not passive: deteriorating health can push a waiting customer to **change class** or to **leave the system** — for alternative care, or through death.

priority · non-priority · jockeying · abandonment

How do jockeying and abandonment interact within a priority discipline — and how does that interaction affect our ability to solve the queue in closed form?

OUTLINE

Five parts

I	The toolbox	M/M/1 · M/M/1+M · generating functions · where derivatives come from
II	Model X and its fundamental equation	the system · balance equations · the fundamental equation · the empty-state theorem
III	The solved models	one theorem per model: A · B_2 · C_2 · B^{H} · C^{H} — and why B stays open
IV	The $\tilde{S}$ experiment	an alternative state space, and a documented negative result
V	Results	CTMC validation · jockeying vs. abandonment, across load · convergence to the baseline

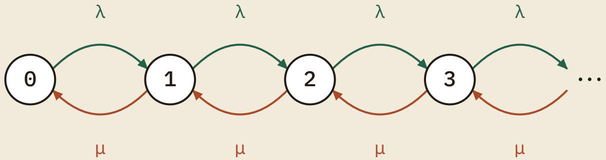
Part I — Preliminaries

The toolbox

Two classical queues, one transform, and the mechanic that turns waiting-room behaviour into derivatives.

The M/M/1 queue: one chain, three formulas

BIRTH–DEATH CHAIN — ARRIVALS Λ UP, DEPARTURES Μ DOWN



DETAILED BALANCE ⇒ GEOMETRIC LAW

$$\lambda\,\pi_n = \mu\,\pi_{n+1} \quad\Longrightarrow\quad \pi_n = (1-\rho)\,\rho^n, \qquad \rho = \tfrac{\lambda}{\mu} < 1$$

MEAN QUEUE AND SOJOURN (LITTLE)

$$\mathbb{E}[L] = \frac{\rho}{1-\rho}, \qquad \mathbb{E}[W] = \frac{1/\mu}{1-\rho}$$

BUSY PERIOD — REMEMBER THIS ONE

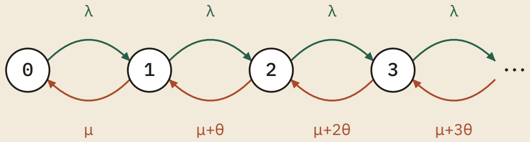
$$\widetilde{BP}(s) = \frac{(\mu+\lambda+s) - \sqrt{(\mu+\lambda+s)^2 - 4\mu\lambda}}{2\lambda}$$

$$\mathbb{E}[BP] = \frac{1}{\mu(1-\rho)}$$

The busy period returns later as the ***effective service time*** *a class-2 customer experiences: the server must first drain the class-1 queue.*

M/M/1+M: impatience brings in Kummer

IMPATIENCE CHAIN — DEPARTURES GROW WITH THE QUEUE: M, M+Θ, M+2Θ, …



WAITING CUSTOMERS ABANDON AT RATE Θ EACH

$$\lambda\,\pi_n = (\mu + n\theta)\,\pi_{n+1} \implies \pi_n = \pi_0\,\frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}}$$

WHAT WE SOLVE FOR — IDLE PROBABILITY & QUEUE-LENGTH PGF

$$\pi_0 = \left[{}_1F_1\!\left(1; \tfrac{\mu}{\theta}; \tfrac{\lambda}{\theta}\right)\right]^{-1}, \quad \Pi(z) = \frac{{}_1F_1\!\left(1; \frac{\mu}{\theta}; \frac{\lambda}{\theta}z\right)}{{}_1F_1\!\left(1; \frac{\mu}{\theta}; \frac{\lambda}{\theta}\right)}$$

*Stable for **every** load — the departure rate $\mu+n\theta$ grows without bound, so no $\rho<1$ is needed.*

THE KUMMER FUNCTION ${}_1F_1(A; B; Z)$

The confluent hypergeometric function — the entire solution of Kummer's equation $z\,u''+(b-z)u'-a\,u = 0$ that is regular at $z = 0$.

POWER SERIES · $(A)_n$ THE RISING FACTORIAL

$${}_1F_1(a;b;z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n}\frac{z^n}{n!}, \qquad (a)_n = a(a+1)\cdots(a+n-1)$$

EULER INTEGRAL FORM · RE B > RE A > 0

$${}_1F_1(a;b;z) = \frac{1}{B(a,b-a)}\int_0^1 e^{zt}\,t^{a-1}(1-t)^{b-a-1}\,dt$$

The integral form is the one that returns: Models B2 and C2 recover P_y as exactly this Beta-weighted integral of e^{zt}.

One full transform, one partial

THE BIVARIATE PGF — THE CENTRAL OBJECT

$$P(x,y) = \mathbb{E}\left[x^{N_1} y^{N_2}\right]$$

$$= \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2)\, x^{n_1} y^{n_2}$$

BOUNDARY FUNCTIONS (EMPTY-QUEUE TRACES)

$$P_x(x) = P(x,0), \qquad P_y(y) = P(0,y)$$

THE PARTIAL PGF (COHEN, 1956)

$$\widetilde{P}(n_1, y) = \mathbb{E}\left[y^{N_2} 1_{\{N_1 = n_1\}}\right]$$

$$P(x,y) = \sum_{n_1} \widetilde{P}(n_1, y)\, x^{n_1}$$

Transform only the hard direction, keep the easy class discrete: the priority class is often a plain M/M/1, so spend the analytical effort on class 2 alone. This idea powers Part IV.

Where the derivatives come from

Jockeying and abandonment act on **every waiting customer**, so their aggregate rate is length-proportional: $\gamma_1 n_1$, $\theta_1 n_1$. A factor n_1 under the transform is precisely a derivative:

$$\sum_{n_1,n_2} n_1\,\pi(n_1,n_2)\,x^{n_1}y^{n_2} \;=\; x\,\frac{\partial P}{\partial x}(x,y)$$

- rate $\gamma_1 n_1$ or $\theta_1 n_1$ → derivative term → ODE/PDE
- constant rate $\gamma_1 \cdot 1\{n_1 \geq 1\}$ (head-of-line) → algebraic term γ_1[P−P_y]

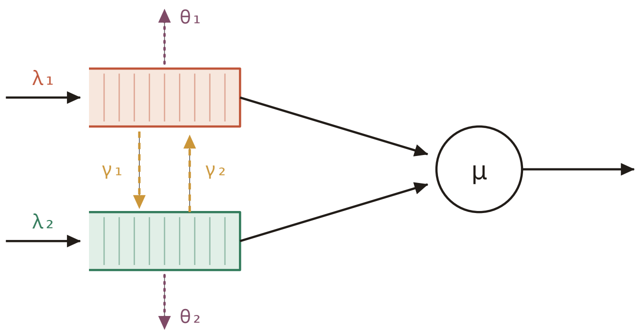
This single mechanic decides the difficulty of every model in the thesis.

Part II — The general model

Model X and its fundamental equation

One equation carries the whole thesis: kernel, two convection terms, two unknowns.

Two classes, one server, both mechanisms



non-preemptive priority · $\gamma_i,\ \theta_i \geq 0$

Poisson arrivals λ_1, λ_2; common exponential service μ; the server never interrupts a service in progress. Every **waiting** customer carries independent exponential jockeying and patience clocks.

jockeying γ_1, γ_2 | abandonment θ_1, θ_2 | $\rho = \lambda/\mu$

n_1, n_2 count customers ***waiting*** *— the one in service is excluded. Its class never matters: memorylessness makes the residual service*

Exp(μ) either way.

One system, two coordinate systems

REGULAR SPACE S — USED THROUGHOUT

$$S = \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}$$

ALTERNATIVE SPACE S̃ — TRACK CLASS 2 AND THE TOTAL

$$\tilde{S} = \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}$$

$$n = n_1 + n_2$$

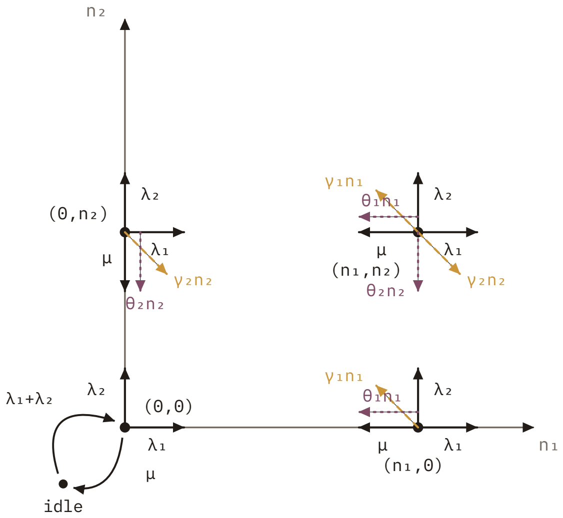
Why S̃? Under jockeying neither class is simple on its own — but the ***total*** *is: jockeying conserves it. Part IV tests whether that helps.*

EMPTY STATES — EXACT FOR THE FULL MODEL X

$$\pi(0,0) = \rho\,\pi_0, \qquad \pi_0 = \frac{1}{1 + \lambda\,\mathbb{E}[B]}$$

A cut between idle and busy, plus renewal–reward over busy cycles: everything reduces to one scalar, the mean busy period **E**[**B**]. Without abandonment, $E[B]=(\mu-\lambda)^{-1}$ gives $\pi_0=1-\rho$, $\pi(0,0)=\rho(1-\rho)$.

The flow at a generic state



INTERIOR STATES ($N_1 \geq 1$, $N_2 \geq 1$)

$$\left[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n_1 + (\gamma_2 + \theta_2)n_2\right] \pi(n_1, n_2)$$
$$= \mu\, \pi(n_1{+}1, n_2) + \lambda_1\, \pi(n_1{-}1, n_2) + \lambda_2\, \pi(n_1, n_2{-}1)$$
$$+\gamma_1(n_1{+}1)\, \pi(n_1{+}1, n_2{-}1) + \gamma_2(n_2{+}1)\, \pi(n_1{-}1, n_2{+}1)$$
$$+\theta_1(n_1{+}1)\, \pi(n_1{+}1, n_2) + \theta_2(n_2{+}1)\, \pi(n_1, n_2{+}1)$$

Plus three boundary families ($n_1=0$; $n_2=0$; the empty state) and the idle cut $(\lambda_1+\lambda_2)\pi_0 = \mu\pi(0,0)$. The length-proportional rates $\boldsymbol{\gamma_i n_i}$, $\boldsymbol{\theta_i n_i}$ *are what become derivatives.*

The fundamental equation of Model X

$$\Big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\Big]\, P(x,y)$$
$$+\; xy\Big[\gamma_1(x-y) + \theta_1(x-1)\Big]\frac{\partial P}{\partial x} \;+\; xy\Big[\gamma_2(y-x) + \theta_2(y-1)\Big]\frac{\partial P}{\partial y}$$
$$=\; \mu(x-y)\, P_y(y) \;-\; \mu x(1-y)\, \pi(0,0)$$

- kernel — exactly Model A's algebraic coefficient
- jockeying factors vanish on the diagonal x=y
- abandonment factors vanish at x=1 (resp. y=1)
- the two unknowns

The vanishing factors choose the technique

Model A · no convection

`algebraic kernel`

No derivative terms. The kernel vanishes on a curve; evaluate at the root **x*(y)** inside the unit disc.

Model B_2 · γ_1 only

`(x−y) → interior singularity`

The singularity sits at the diagonal point **x=y**, with negative exponent: **analyticity** there pins P_y(y).

Model C_2 · θ_1 only

`(x−1) → boundary singularity`

The singularity moves to **x=1** on the unit circle, with positive exponent: mere **boundedness** suffices.

Model B^H **Model C^H**

`head-of-line → collapse`

Constant rate $1\{n_1 \geq 1\}$ removes the derivative outright — back to Model A's algebraic form.

Theorem 4.1 — two identities pin the empty states

THEOREM 4.1 *empty-state probabilities of the general model*

Before any model is solved: for the full Model X, with both mechanisms active,

$$\pi(0,0) = \rho\,\pi_0$$

$$\pi_0 = \frac{1}{1+\lambda\,\mathbb{E}[B]}, \qquad \pi(0,0) = \frac{\rho}{1+\lambda\,\mathbb{E}[B]}$$

Proof. flux cut $\lambda\pi_0 = \mu\pi(0,0)$ between the idle and busy states · renewal–reward over regeneration cycles at the empty state — exact for every (γ_i, θ_i). Everything reduces to one scalar, E[B].

MODEL PRIMITIVES

λ_1, λ_2	Poisson arrival rates — **class 1** has non-preemptive priority over **class 2**
μ	exponential service rate, common to both classes
λ, ρ	$\lambda = \lambda_1 + \lambda_2$ total arrival rate; $\rho = \lambda/\mu$ the offered load
γ_i, θ_i	jockeying / abandonment rates — **arbitrary ≥ 0**: the theorem holds for the whole family

OBJECTS IN THE STATEMENT

π_0	stationary probability of the **empty system** — server idle, nobody waiting
π(0,0)	server **busy**, both queues empty: $N_1 = N_2 = 0$ waiting
E[B]	mean **busy period**: from an arrival into an empty system until the system next empties
assumption	the CTMC of Model X is positive recurrent, so the

▸ **Corollary 4.4** *No abandonment ⇒ the M/M/1 baseline, for A, B, B2 and B^H at once* EXPAND

Part III — Analysis

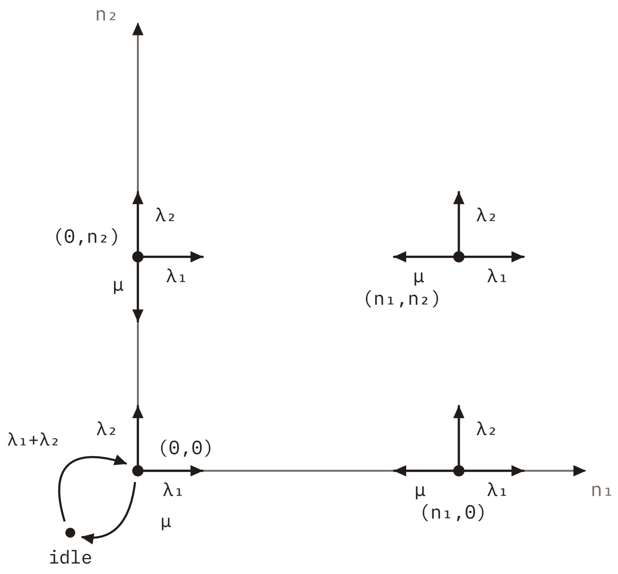
The solved models — and the one that is not

Baseline A, the open Model B, one-way jockeying B2, abandonment C2, and the head-of-line pair.

Seven models, one functional equation

MODEL	ACTIVE MECHANISM	EQUATION TYPE	STATUS
X	jockeying & abandonment	PDE · ???	• Model X
A	none — baseline	algebraic PGF	• Model A
B	bidirectional jockeying	PDE · Volterra family	• Model B
B_2	one-way jockeying (1→2), $\gamma_1 n_1$	ODE in x · Kummer	• Model B_2
C_2	class-1 abandonment, $\theta_1 n_1$	ODE in x · Kummer	• Model C_2
B^H	jockeying, head-of-line $\gamma_1 \cdot 1\{n_1 \geq 1\}$	algebraic PGF	• Model B^H
C^H	abandonment, head-of-line $\theta_1 \cdot 1\{n_1 \geq 1\}$	algebraic PGF	• Model C^H

No mechanisms: the equation is algebraic



REDUCED FUNDAMENTAL EQUATION

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y)$$
$$= \mu(x-y)P_y(y) - \mu x(1-y)\pi(0,0)$$

Fix y and find the root $\mathbf{x}^*(\mathbf{y})$ of the kernel inside the unit disc. P is bounded there, so the LHS vanishes — hence the RHS must too:

$$P_y(y) = \frac{x^*(y)\,(1-y)}{x^*(y)-y}\,\pi(0,0)$$

THE EMPTY STATE — FOR COMPLETENESS

$$\pi(0,0) = (1-\rho)\,\rho$$

Theorem 5.1 — the baseline PGF

THEOREM 5.1 *closed-form PGF of Model A*

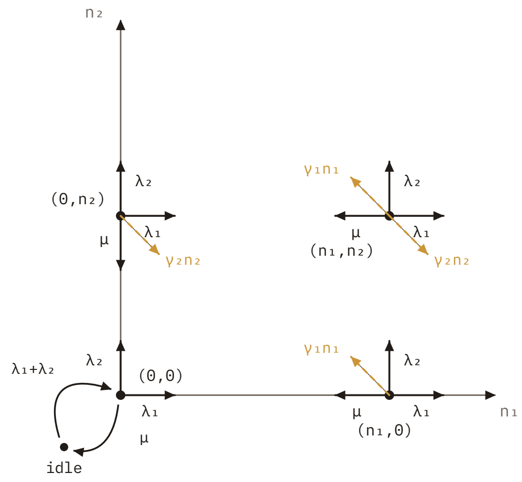
Under ρ < 1, the joint PGF is

$$P(x,y) = \frac{\rho(1-\rho)\,y(1-y)}{(1{+}\rho_1{+}\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2} \cdot \frac{x - x^*(y)}{x^*(y) - y}$$

$$x^*(y) = \frac{\left[\mu + \lambda_1 + \lambda_2(1-y)\right] - \sqrt{\left[\mu + \lambda_1 + \lambda_2(1-y)\right]^2 - 4\lambda_1\mu}}{2\lambda_1}$$

Proved twice. kernel method (analytic) · Poisson splitting + PASTA (probabilistic) — the two routes meet in the same root x*(y). Classical result, recovered as calibration for everything that follows.

Now both derivatives survive



MODEL-B FUNDAMENTAL EQUATION

$$\begin{aligned}&\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P \\ &+ \gamma_1 xy(x-y)\frac{\partial P}{\partial x} + \gamma_2 xy(y-x)\frac{\partial P}{\partial y} \\ &= \mu(x-y)P_y(y) - \mu x(1-y)\pi(0,0)\end{aligned}$$

Both convection terms carry (x−y): the Method of Characteristics applies, along the straight lines

$$C_1 = \gamma_2 x + \gamma_1 y$$

And jockeying destroys the Poisson split of arrivals — the probabilistic route is gone too.

Model B reduces to a Volterra-type integral equation

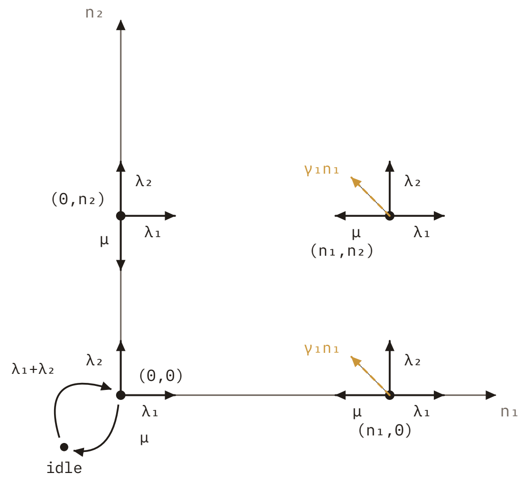
CLAIM 6.3 *integral representation along characteristics*

Along each characteristic line, variation of parameters gives P = P_h · ∫ h/P_h. Evaluating on the diagonal x=y=z — where P(z,z)=π(0,0)/(1−ρz) is already known — the boundary function must satisfy a **Volterra integral equation of the first kind**:

$$\mathcal{F}(z) = \int_0^z \mathcal{K}(z,t)\; P_y\big(y(t;z)\big)\, dt$$

The forcing depends on the unknown boundary itself. Inverting it needs tools beyond this thesis — **Model B is left open**, and it is the representative of every model with a ∂/∂y term, Model X included.

Kill the ∂/∂y term, keep the jockeying



REDUCED EQUATION — AN ODE IN X FOR EACH FIXED Y

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P + \gamma_1 xy(x-y)\frac{\partial P}{\partial x}$$
$$= \mu(x-y)P_y(y) - \mu x(1-y)\pi(0,0)$$

The integrating factor has a singularity at the **interior** point x=y with negative exponent β(y). A PGF must be analytic there — that single demand determines P_y(y).

$\alpha(y) = \mu/(\gamma_1 y)$ $\beta(y) = (1-y)(\lambda y-\mu)/(\gamma_1 y)$

Theorem 7.1 — the B2 generating function

THEOREM 7.1 *boundary function and PGF of Model B2*

Under ρ < 1, analyticity of P at the interior point x = y forces

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu - \lambda y} \cdot \frac{{}_1F_1\big(\alpha(y)+1;\ b^\star(y);\ -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\ b^\star(y);\ -\lambda_1 y/\gamma_1\big)}$$

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}}{\gamma_1\, y\, x^{\alpha(y)}(y-x)^{\beta(y)}}\Big[P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\,\pi(0,0)\,\mathcal{H}_2(x,y)\Big]$$

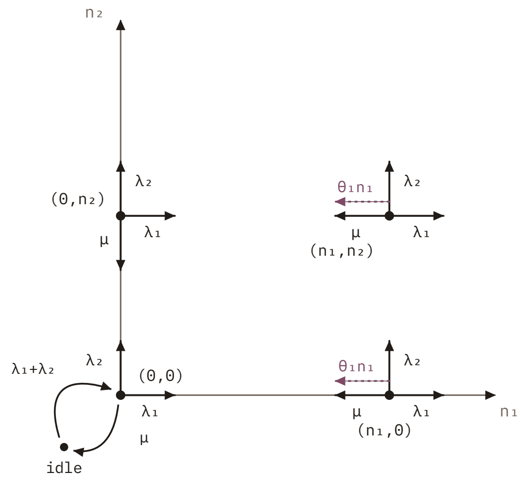
valid on 0 ≤ x ≤ y; extends by continuation across x = y.

THE HELPER INTEGRALS

$$\mathcal{H}_1(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\ t^{\alpha(y)-1}\,(y-t)^{\beta(y)}\,dt \qquad \mathcal{H}_2(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\ t^{\alpha(y)}\,(y-t)^{\beta(y)-1}\,dt$$

$\mathcal{H}_1$ weighs P_y(y); $\mathcal{H}_2$ weighs (1−y)π(0,0) — exponents traded between t = 0 and t = y.

Abandonment moves the singularity to x = 1



REDUCED EQUATION

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P + \theta_1 xy(x-1)\frac{\partial P}{\partial x} = \mu(x-y)P_y(y) - \mu x(1-y)\pi(0,0)$$

Queue 1 alone is the **Erlang-A** of Part I; class 2 sees an M/G/1 whose service time is its busy period **B_C**. Stability is genuinely non-trivial:

$$\lambda_2\,\mathbb{E}[B_C] < 1 \iff \rho_2 \cdot {}_1F_1\left(1; \tfrac{\mu}{\theta_1}+1; \tfrac{\lambda_1}{\theta_1}\right) < 1$$

Theorem 8.1 — the C2 generating function

THEOREM 8.1 *PGF of Model C2, pinned by boundedness at x = 1*

Under λ2E[B_C] < 1, writing B̃_C for B̃_C(λ2(1−y)),

$$P(x,y) = \frac{\mu\,\pi(0,0)(1-y)\,e^{\lambda_1 x/\theta_1}}{\theta_1\, x^{\mu/\theta_1}(1-x)^{\lambda_2(1-y)/\theta_1}\left(\widetilde{B}_C - y\right)}\left[\widetilde{B}_C\,\mathcal{H}_1(x,y) - \left(1-\widetilde{B}_C\right)\mathcal{H}_2(x,y)\right]$$

THE HELPER INTEGRALS AND THE BUSY-PERIOD LST

$$\mathcal{H}_1(x,y) = \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1 - 1}\,(1-t)^{\lambda_2(1-y)/\theta_1}\,dt \qquad \mathcal{H}_2(x,y) = \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1}\,(1-t)^{\lambda_2(1-y)/\theta_1 - 1}\,dt$$

$$\widetilde{B}_C(s) = \mathbb{E}\left[e^{-sB_C}\right] = \frac{\mathrm{B}(a+1,b)\,{}_1F_1(a+1;a+1+b;-c)}{\mathrm{B}(a,b)\,{}_1F_1(a;a+b;-c)}$$

with $a = \mu/\theta_1$, $b = s/\theta_1$, $c = \lambda_1/\theta_1$, and $s = \lambda_2(1-y)$.

Restrict to the head of the line — the derivative disappears

Let only the customer at the **head of Queue 1** jockey or abandon: the rate becomes the constant $\gamma_1 \cdot 1\{n_1 \geq 1\}$ (resp. $\theta_1 \cdot 1\{n_1 \geq 1\}$), and the mechanism joins the kernel as an algebraic term:

B^H — JOCKEYING JOINS THE KERNEL

$$\left[\text{kernel} + \gamma_1 y(x-y)\right] P$$
$$= \left[\mu(x-y) + \gamma_1 y(x-y)\right] P_y - \mu x(1-y)\pi(0,0)$$

C^H — ABANDONMENT JOINS THE KERNEL

$$\left[\text{kernel} + \theta_1 y(x-1)\right] P$$
$$= \left[\mu(x-y) + \theta_1 y(x-1)\right] P_y - \mu x(1-y)\pi(0,0)$$

Both are solved by the Kernel Method exactly as Model A. For C^H even stability relaxes: $(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1 > 0$ — weaker than $\rho < 1$.

Theorem 9.1 — head-of-line jockeying is rational

THEOREM 9.1 *closed-form PGF of Model B^H*

Under ρ < 1, the joint PGF is the rational function

$$P(x,y) = \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\big(x^*(y)-y\big)\big(x^+(y)-x\big)}$$

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big]x + \big(\mu + \gamma_1 y\big) = 0$$

Proof. the constant rate $\gamma_1 \cdot 1\{n_1 \ge 1\}$ joins the kernel as an algebraic term — no derivative survives, and the Kernel Method runs exactly as for Model A. CTMC check: max error < 3×10^{-7} on a 5×5 grid. As $\gamma_1 \to 0^+$ the formula collapses to Theorem 5.1.

EVERY SYMBOL IN THE STATEMENT

γ_1	**head-of-line** jockeying: only the first waiting class-1 customer may switch, so the aggregate rate is the constant $\gamma_1 \cdot 1\{n_1 \ge 1\}$ ($\gamma_2 = \theta_1 = \theta_2 = 0$)
λ, ρ	$\lambda = \lambda_1+\lambda_2$, $\rho = \lambda/\mu < 1$ — stability; the total count is still conserved
$P(x,y)$	joint PGF E[x^N₁ y^N₂] of the waiting counts, over busy states
$\rho(1-\rho)$	the numerator constant = π(0,0), by Corollary 9.2 below
$x^*(y)$, $x^+(y)$	the two roots of the kernel quadratic on the left — **x* inside** the unit disc (where boundedness pins the RHS), **x^+ outside**
$\mu + \gamma_1 y$	the constant term of the quadratic: service plus the jockey stream re-entering through Queue 2

Theorem 10.1 — head-of-line abandonment stays rational

THEOREM 10.1 *closed-form PGF of Model C^H*

Under $(\mu+\theta_1)(\mu-\lambda) + \lambda_1\theta_1 > 0$, the joint PGF is the rational function

$$P(x,y) = \frac{\mu(\mu+\theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\big(x^+(y)-x\big)\big[(\mu+\theta_1 y)\,x^*(y) - y(\mu+\theta_1)\big]}$$

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \theta_1\big]x + \big(\mu+\theta_1\big) = 0$$

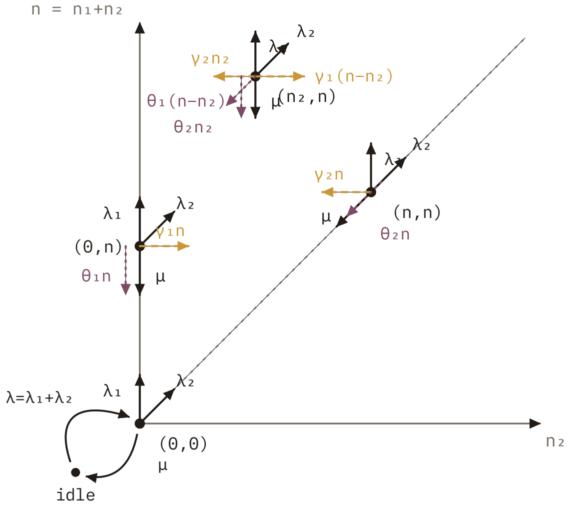
Proof. Kernel Method again — abandonment enters the kernel as the algebraic term $\theta_1 y(x-1)$. CTMC check: max error $< 3\times10^{-11}$ on a 5×5 grid. As $\theta_1 \to 0^+$ the formula recovers Theorem 5.1 exactly.

Part IV — A documented negative result

The $\tilde{S}$ experiment

If jockeying conserves the total, why not make the total a coordinate? Because the recurrence refuses to close.

Model X in the (n_2, n) coordinates



INTERIOR STATES ($1 \le N_2 \le N-1$, $N \ge 2$)

$$\left[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)(n-n_2)+(\gamma_2+\theta_2)n_2\right]\tilde{\pi}(n_2,n)$$
$$=\lambda_1\tilde{\pi}(n_2,n-1)+\lambda_2\tilde{\pi}(n_2-1,n-1)+\mu\,\tilde{\pi}(n_2,n+1)$$
$$+\gamma_1(n-n_2+1)\,\tilde{\pi}(n_2-1,n)+\gamma_2(n_2+1)\,\tilde{\pi}(n_2+1,n)$$
$$+\theta_1(n+1-n_2)\,\tilde{\pi}(n_2,n+1)+\theta_2(n_2+1)\,\tilde{\pi}(n_2+1,n+1)$$

Arrivals now move diagonally (λ_2 raises both n_2 and n); jockeying moves horizontally at fixed n; the diagonal $n_2 = n$ means **all waiting customers are class 2** *— it will matter.*

The fundamental equation on S̃

Transform only n2 and keep the total n discrete: P̃(y,n) = Σ π̃(n2,n)y^n2, a polynomial of degree n.

$$\Big[-(\gamma_2+\gamma_1 y)(1-y) + (\theta_2-\theta_1)y\Big]\frac{d}{dy}\widetilde{P}(y,n) + \Big[\lambda_1+\lambda_2+\mu + \gamma_1 n(1-y) + \theta_1 n\Big]\widetilde{P}(y,n)$$

$$= (\lambda_1+\lambda_2 y)\,\widetilde{P}(y,n-1) + \Big[\mu+\theta_1(n+1)\Big]\widetilde{P}(y,n+1) + (\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1)$$

$$+\ \mu\, y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$$

A differential–**difference** equation: levels n−1, n, n+1 are coupled — plus **one forcing term**.

Why the experiment did not work

MODEL A ONLY: DROP THE FORCING (IT DECAYS LIKE P^n)

$$\widetilde{P}(y,n) \approx \rho(1-\rho)\,[\tilde{y}^*(y)]^n, \quad \tilde{y}^*(y) = \frac{(\lambda+\mu) - \sqrt{(\lambda+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}$$

The forcing term $\mu y^n(1-y)\tilde{\pi}(n+1,n+1)$ feeds in the **diagonal** states — all waiting customers class 2. Those probabilities are exactly what we are solving for: the recurrence **does not close on itself**.

Instead of collapsing 2-D balance to a 1-D recurrence, the change of coordinates makes things **harder** — for every model with an active mechanism, $\tilde{S}$ is strictly worse than S.

Even then it is an **approximation, not a theorem** *— accurate only where the forcing decays faster than the solution, $\rho < \tilde{y}^*(y)$: a priority-heavy, moderate load. At $y=1$ it does recover the M/M/1 marginal $(1-\rho)\rho^{n+1}$.*

deliberate negative result

Part V — Results

Validate, then compare

Every closed form against an exact CTMC; then jockeying and abandonment head to head.

Every closed form matches the CTMC

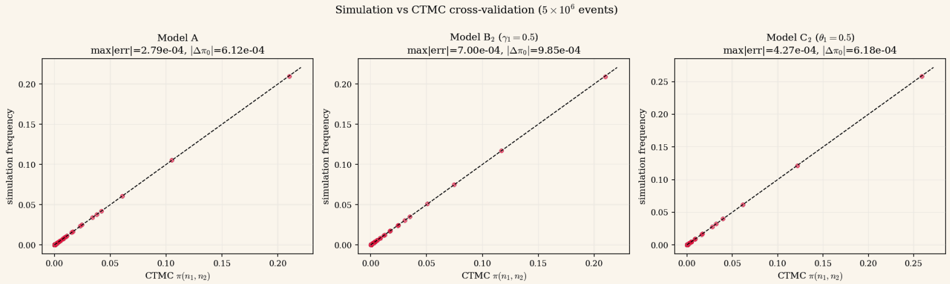


Fig. 1. Analytic PGF marginals against a truncated CTMC that recovers the baseline as the mechanism rates vanish.

`simulation vs. CTMC · all solved models`

CONSERVATION CHECK (C_2)

$<10^{-14}$

offered = carried + lost

BASELINE RECOVERY

exact

mechanism rates → 0 ⇒ Model A

The abandonment gap widens with load

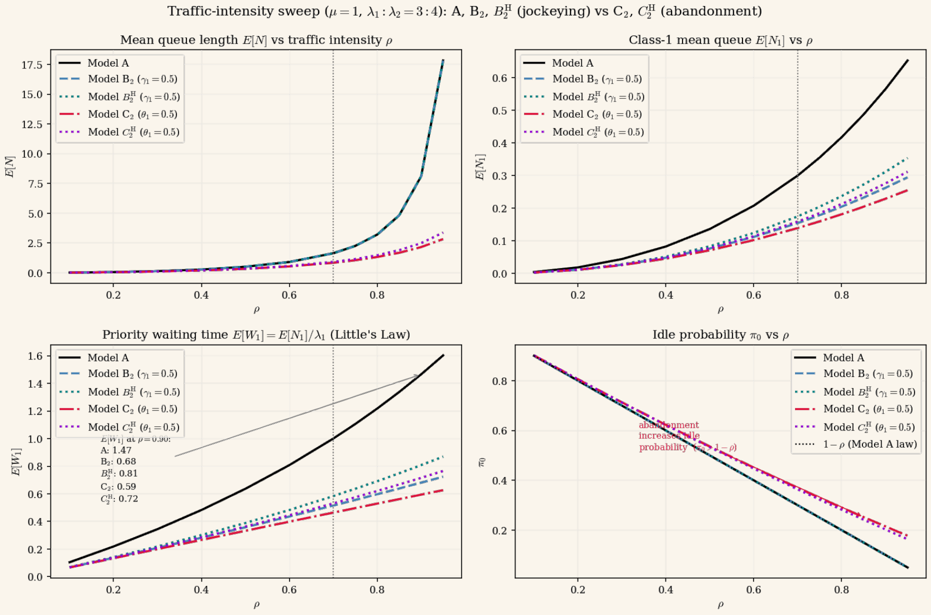


Fig. 4. A, B_2 and B^H coincide in E[N] and obey the $1-\rho$ idle law — jockeying conserves N. C_2 and C^H sit below in E[N] and above in π_0 at every load; the head-of-line pair interpolates.

`five solved models · μ = 1 · λ₁:λ₂ = 3:4 · γ₁ = θ₁ = 0.5`

CLASS-2 QUEUE $E[N_2]$ · P = 0.90

1.925 ↓

vs. 7.535 · Model A

IDLE PROBABILITY Π_0 · P = 0.70

0.370 ↑

vs. 0.300 = $1-\rho$ · A and B_2

CLASS-2 WAIT $E[W_2]$ · B_2

+10.9 % ↑

3.697 vs. 3.333 · jockeying penalises class 2

Five solved models, side by side

	A	B_2	B^H	C_2	C^H
MECHANISM RATE	none	$\gamma_1 n_1$	$\gamma_1 \cdot 1\{n_1 \geq 1\}$	$\theta_1 n_1$	$\theta_1 \cdot 1\{n_1 \geq 1\}$
FUNDAMENTAL EQ.	algebraic	1st-order ODE	algebraic	1st-order ODE	algebraic
PINNING OF P_Y	kernel root x*	analyticity at x=y	kernel root x*	boundedness at x=1	kernel root x*
P(X,Y) FORM	rational	integral + ${}_1F_1$	rational	integral + ${}_1F_1$	rational
CONSERVES N	yes	yes	yes	no	no
Π(0,0)	$\rho(1-\rho)$	$\rho(1-\rho)$	$\rho(1-\rho)$	Cor. 8.2	Cor. 10.2

Head-of-line rates collapse the ODE back to the algebraic form — P(x,y) becomes rational again.

Four mechanisms solve, two obstruct

1. **Kernel root** inside the disc (A) · **analyticity** at x=y (B_2) · **boundedness** at x=1 (C_2) · **head-of-line collapse** (B^H, C^H) — the four ways P_y(y) gets pinned.
2. The obstructions: a **∂/∂y term** forces a first-kind Volterra equation (B); Model X adds transcendental characteristics on top.
3. **Jockeying redistributes** delay at fixed total work; **abandonment reduces** the load itself. Same waiting room, opposite registers.

What the machinery makes possible next

Head-of-line ladder

Let the first k customers act: $\min(n_1,k)$ keeps the equation algebraic at the price of k boundary unknowns; expect convergence to B_2 / C_2 as $k \to \infty$.

Invert the Volterra equation

Model B's kernel and forcing are explicit — numerical inversion (or sharper analysis) would close the first obstruction.

Exact means from the PGFs

Differentiate at (1,1) — with L'Hôpital at the x=1 singularity — to replace the numerical $E[N_i]$, $E[W_i]$ with closed forms.

The clinically wanted direction

The 2→1 upgrade of a deteriorating patient is exactly $\gamma_2 > 0$ — the Volterra-hard direction. Calibration to elderly-care data comes after.

THANK YOU

Questions welcome.

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